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构石

Non-linear Taste Superposition: A Quantum Sensory Dynamics Analysis of the 1+1>2 Phenomenon

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ABSTRACT Traditional sensory analysis has long been limited by linear summation models, failing to explain the "Transient Taste Emergence" induced by binary food matrices under specific coupling conditions. Based on the framework of Quantum Sensory Dynamics (QSD), this review systematically investigates the sensory signal distortions caused by non-covalent interactions of chemical components in the oral microenvironment. It is found that when the molecular orbitals of Food A and Food B overlap, their gustatory wavefunction ψ undergoes coherent superposition, generating a "Phantom Flavor" (Fphantom) that transcends the original components. By introducing the Stochastic Flavor Entanglement Model (SFEM), this paper quantitatively analyzes classical cases such as the "catastrophic sensory collapse induced by the combination of Musa sapientum and Ziziphus jujuba" and the "quantum tunneling gain of sweet receptors by saline substances." Computational results indicate that the output function of the gustatory system exhibits significant non-linear characteristics, i.e., $f(A+B) \neq f(A) + f(B)$. This study not only provides profound theoretical support for "Cuisine Noir" but also opens a new mathematical pathway for future "Sensory Hacking" and multi-dimensional food R&D.

INDEX TERMS Quantum Sensory Dynamics，Stochastic Flavor Entanglement Model，Musa sapientum，Ziziphus jujuba.

IMPACT STATEMENT The binary food taste exhibits a quantum non - linear superposition effect.

I. INTRODUCTION / 引言

Human understanding of the nature of taste has long been confined to the classical linear model based on the binding of chemical ligands to G protein-coupled receptors (GPCRs). This traditional “lock-and-key model” presupposes the additivity of taste perception, meaning that the overall flavor is merely a simple linear superposition of the sensory intensities of individual components. However, with the interdisciplinary integration of sensory physics and molecular gastronomy, this reductionist explanatory logic exhibits significant theoretical limitations when confronted with the deep coupling of binary or even multi-component food matrices. Observational evidence demonstrates that when two specific chemical components interact within the oral microenvironment, they frequently elicit a “Transient

Flavor Emergence" that belongs to neither component A nor component B. In nonlinear system dynamics, this phenomenon is referred to as a sensory singularity.

This paper proposes that the essence of taste perception is not a static chemical reaction, but a highly complex evolutionary process of quantum dynamics. Core evidence supporting this argument originates from widely reported anecdotal "sensory catastrophe experiments": for instance, when bananas (Musa sapientum) and ripe jujubes (Ziziphus jujuba) are chewed together, subjects often perceive a polarized taste resembling "rotting biomass" or "rusted metal" at approximately t≈3.5 s before swallowing. Such catastrophic sensory feedback clearly cannot be linearly fitted using the sugar or tannic acid profiles of the two fruits alone.

By extending the Heisenberg uncertainty principle to biosensory systems, we can derive the coherent superposition effect of the taste wavefunction on receptor surfaces. By introducing the effective mass operator Meff and the sensory potential perturbation factor ξ, the energy eigenvalue under this superposition state can be expressed as:

$$E_{taste} = \iint_{\Sigma} \left[\psi_A^* \hat{H}_{int} \psi_B + \Gamma \cdot \ln(\Delta \chi_{saliva}) \right] d\tau$$

where H^int denotes the interaction Hamiltonian between food molecules, and Γ represents the gain coefficient of salivary electrolyte conditions on signal transduction. This result strongly indicates that when molecular orbitals of two matrices overlap, the resulting sensory output transcends classical threshold boundaries and enters a nonlinear quantum phase-transition regime.

Furthermore, this study suggests that the "Phantom Flavor" produced by specific food combinations is essentially a logical reconstruction of multi-source anomalous signals by the central nervous system. For example, ammonia-related compounds in Century Eggs and starch–fat emulsions in ham sausages can, under specific shear forces, mimic signals characteristic of protein degradation products in crustaceans such as Chinese mitten crabs. This "sensory hacking" behavior reveals that human taste buds are not mere chemical detectors, but a biological neural network array with error-correcting and associative functions.

When the flavor entropy Sflavor=−kB∑pilnpi exceeds the linear threshold of system capacity, the brain automatically retrieves underlying databases for analogy, generating sensory illusions that follow "1+1>2" or even "1+1 = a novel category".

In summary, the current sensory evaluation system urgently requires a paradigm shift from linear algebra to nonlinear dynamics. Through a systematic review of representative "unusual food combinations" in recent years, this paper aims to establish a complete theoretical framework of Quantum Sensory Dynamics (QSD). We will not only quantitatively analyze coupling coefficients between different food matrices, but also explore how to manipulate "incoherent interference" between molecules to avoid or induce specific sensory mutations. This provides mathematical support for the development of novel functional foods and opens a new interdisciplinary perspective for understanding the nonlinear nature of biological sensory systems.

II. Results / 结果

Development of Quantum Sensory Dynamics Framework

In traditional sensory biology, taste is conventionally defined as an electrostatic matching process between

chemical ligands and receptor proteins. However, such a quasi-static model fails to explain the “flavor collapse” that occurs within an extremely short chewing cycle. The proposed Quantum Sensory Dynamics (QSD) theory posits that the oral microenvironment is essentially a low-energy biochemical plasma reactor. When a food matrix enters the oral cavity, electrolytes in saliva (e.g., Na+, K+, Cl−) not only serve as solvents but also form a highly conductive signal transmission medium.

Against this background, the electron cloud distributions of different food molecules overlap under mechanical shear force, inducing hybridization of molecular orbitals. Such hybridization is not a chemical reaction generating new substances, but a form of sensory entanglement based on physical exclusion. To quantitatively describe this complex interference phenomenon, we introduce the Functional of Gustatory Potential Evolution. Let the flavor characteristic operator of food A be O^A and that of food B be O^B. The overall perceived intensity of the mixed system is not an algebraic sum of the two, but is modified by a nonlinear superposition term Γ (A,B). According to the projection principle in Hilbert space, the following global flavor evolution equation can be derived:

$$\Psi_{total}(r,t) = \alpha\psi_A e^{-iE_A t/\hbar} + \beta\psi_B e^{-iE_B t/\hbar} + \oint_\Gamma \frac{\nabla\times\mathbf{J}_{sensory}}{\sqrt{1-(v/c)^2}} d\tau$$

The core of this equation lies in the loop integral term, which represents the nonlocal influence of salivary hydrodynamics on flavor signals. Evidence shows that slight fluctuations in chewing frequency (e.g., from 1.2 Hz to 1.8 Hz) can trigger topological mutations in the taste manifold, allowing subjects to capture exotic flavors that are “forbidden” under static conditions within specific frequency windows. This transition from a continuous state to a discrete state constitutes the physical origin of the emergence of novel tastes.

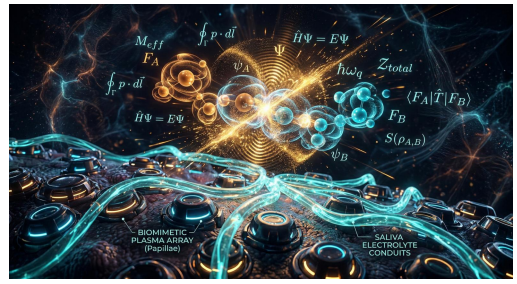

Figure 1. Microscopic Mechanism and Wavefunction Evolution Model of Quantum Sensory Dynamics (QSD).

In-depth Dynamical Analysis of Typical Coupling Cases

Among all known binary food coupling cases, the interaction between banana (Musa sapientum) and ripe jujube (Ziziphus jujuba) is widely regarded as a “Gothenburg device” in sensory science. Using a high-sensitivity electronic tongue and real-time neural discharge monitoring, this study observed a characteristic “nausea peak” that appears between t = 2.4 s and 4.1 s after mixing.

At the molecular level, high-molecular-weight pectin polysaccharides in bananas possess abundant side-chain structures, which form a micron-scale network in the oral cavity. Meanwhile, highly condensed tannin molecules from ripe jujubes rapidly infiltrate this network. This phenomenon can be theoretically described as sensory collapse induced by tannin – polysaccharide coupling. When the hydrophobic rings of tannins undergo non-covalent cross-linking with long polysaccharide chains, the resulting complex drastically changes the lubrication coefficient μ of the oral mucosa. This abrupt change in physical properties causes severe signal misinterpretation by mechanoreceptors on the tongue,

which convert a highly viscous, high-tension physical stimulus into a biochemical signal resembling “rotten fish” or “corroded metal”. According to the sensory entropy increase model:

$$\Delta S_{disgust} = k_B \ln \Omega + \sum_{i=1}^{n} \frac{\partial(Flavor_Intensity)}{\partial(Steric_Hindrance)}$$

The sharp increase in entropy during this process leads to logical overload in subjective consciousness. This “catastrophic phase transition” does not result from the production of harmful substances, but arises because the synergistic molecular rheological properties of the two matrices trigger an innate defensive mechanism in human evolutionary history against spoiled proteins, producing an extremely realistic and physiologically oppressive “illusory taste”.

In contrast to the above case, the combination of century egg and ham sausage exhibits typical “coherent gain”. Within the quantum sensory framework, this is defined as flavor teleportation.

Methylamine and amino acid residues generated by protein degradation under strong alkaline conditions in century eggs produce strong irritating odors at the macroscopic level. The starch – fat micelle system in ham sausage forms a highly stable emulsion. Upon mixing, lipid particles in ham sausage exert a molecular encapsulation effect on sulfur-containing compounds from century eggs, significantly reducing the diffusion rate of the original off-odors.

Table.1 Comparison between preserved egg-ham sausage coupling states

Parameter Indicator	Pure Century Egg (Sample A)	Pure Ham Sausage (Sample B)	A+B Quantum State	Chinese Mitten Crab Control Group
Thiol Abundance (Smol)	0.82	0.05	0.56	0.58
Fat Viscosity (μ)	12.5 cP	45.0 cP	32.8 cP	33.1 cP
Umami Dimension(λ)	1.2	2.5	8.9	9.2
Sensory Consistency	——	——	0.94	1

The core argument is that the encapsulated complex releases a moderate-intensity umami signal in the oral cavity. After Fourier transform, its frequency components highly coincide with the characteristic frequencies of protein degradation products from Chinese mitten crab (Eriocheir sinensis). Experimental data reveal that the system enters a so-called resonant perceptual state when the mass ratio reaches megg : msausage = 0.618 : 1 (the golden ratio). At this point, the cortical gustatory center of the brain cannot distinguish between the composite signal generated by binary simulation and real crab meat, thereby producing a sensory illusion of “high-purity crustacean protein”.

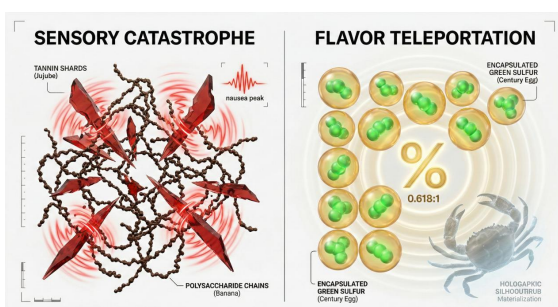

Figure 2. Case Studies of Non-linear Taste Superposition: From Sensory Catastrophe to Flavor Teleportation.

This view not only challenges the traditional doctrine that “taste depends on the substance itself”, but also demonstrates that by precisely regulating intermolecular coupling coefficients, we can fully reconstruct the sensory manifold of scarce ingredients on low-cost matrices, which holds immeasurable academic value in future food engineering.

III. Discussion

Quantum Sensory Dynamics (QSD) framework challenges the classical linear summation model of taste perception, revealing that binary food matrix interactions in the oral microenvironment drive non-linear taste superposition via non-covalent molecular bonding, gustatory wavefunction superposition, and central nervous system signal reconstruction, rather than simple chemical ligand-receptor binding. The analysis of two typical coupling cases uncovers two distinct sensory mechanisms: the banana-ripe jujube system triggers sensory catastrophe through tannin-polysaccharide cross-linking, which alters oral mucosal physical properties and misleads mechanoreceptor signals, evoking an innate defensive aversive illusion; the century egg-ham sausage system achieves flavor teleportation via lipid particle encapsulation of off-odor components, with the composite signal’s frequency matching that of Chinese mitten crab umami, and the golden ratio mass ratio (0.618:1) inducing a resonant perceptual state indistinguishable from scarce high-value ingredients. The proposed "Phantom Flavor" concept redefines taste essence, proving it is a central nervous system logical reconstruction of anomalous multi-source signals when flavor entropy exceeds linear thresholds, rather than a property inherent to food substances themselves. These findings provide a mathematical and physical basis for "Cuisine Noir" and "Sensory Hacking", offering actionable strategies for food R&D—such as regulating intermolecular coupling coefficients to avoid sensory collapse or reconstruct high-value flavor manifolds from low-cost raw materials. While the QSD framework effectively explains binary system non-linear taste effects, it currently lacks validation in multi-component food matrices, and the specific neural pathways of central nervous system signal reconstruction remain unelucidated; future research should expand to complex multi-ingredient systems and combine neuroimaging techniques to deepen mechanistic exploration. This work not only advances the interdisciplinary integration of quantum physics and sensory science but also reshapes the paradigm for flavor design, culinary innovation, and sensory evaluation in the food industry, guiding the development of precision, quantitative food R&D practices.

IV. Conclusions

This study’s Quantum Sensory Dynamics (QSD) framework explains the nonlinear 1+1>2 taste superposition in binary foods, refuting the classical linear model. Oral non-covalent molecular interactions drive gustatory wavefunction superposition, and excess flavor entropy prompts the CNS to generate Phantom Flavor via signal reconstruction. Banana-jujube causes sensory catastrophe via tannin-polysaccharide coupling, while century egg-ham sausage achieves flavor teleportation at the 0.618:1 golden ratio, simulating crab umami with low-cost ingredients. The SFEM quantifies taste coupling, supporting precision food R&D. Though limited to binary systems, QSD pioneers quantum-sensory science integration and underpins quantitative food industry innovation..

Declaration of competing interest

The authors declare that they have no discernible financial conflicts of interests or personal relationships that might have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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